

CS 7637 Mini-Project 2

Kamran Munawar
kmunawar3@gatech.edu

1 HOW DOES YOUR AGENT WORK? DOES IT USE SOME CONCEPTS COVERED IN OUR COURSE? OR SOME OTHER APPROACH?

The agent first preprocesses the sentence by tokenizing it into individual words and normalizing them. It uses internal knowledge representations consisting of word lists such as verbs, pronouns, articles, prepositions, and names. These word lists were derived from mostcommon.txt file.

After preprocessing the agent performs a simple syntactic analysis of the sentence. It identifies the main verb by scanning the sentence for a word classified as a verb. Once the verb is found the agent assumes a simplified sentence structure in which Words before the verb are likely to form the subject. Words immediately after the verb are potential objects. Words following prepositions form location or relational phrases. The agent also examines the question to determine the type of information being requested. “Who” questions typically require the subject or a person-related noun. “What” questions usually require an object or noun phrase. “Where” questions are answered using prepositional phrases indicating location. “When” questions search for temporal expressions in the sentence. Based on the question type, the agent selects the relevant component extracted from the sentence.

The agent implicitly uses a simplified frame like structure when analyzing sentences. Each sentence is interpreted as an event consisting of several components such as:

- **Subject** – the entity performing the action
- **Verb** – the action occurring in the sentence
- **Object** – the entity receiving the action
- **Prepositional phrases** – additional relational or location information

this can be viewed as a simple event frame:

Slot	Example Value
Agent (Subject)	Ada
Action (Verb)	brought
Object	note
Recipient	Irene

For example the sentence “*Ada brought a note to Irene.*” can be interpreted using this frame structure. When the question asks “*Who brought the note?*”, the agent returns the value stored in the *Agent* slot.

The agent uses structured word lists and dictionaries to represent linguistic knowledge such as verbs, names, and prepositions. This corresponds to the concept of knowledge representation discussed in lesson Semantic Networks.

The agent applies explicit rules to map question types to sentence components. For example returning the subject for “Who” questions. This approach is similar to reasoning using production systems discussed in lesson Production Systems.

The agent also relies on heuristics to simplify sentence analysis. For example, assuming that the subject appears before the verb and the object appears after the verb.

2 HOW WELL DOES YOUR AGENT PERFORM? DOES IT STRUGGLE ON ANY PARTICULAR CASES?

Agent achieved a score of 38 out of 40 on the Gradescope. The agent performs well on most simple sentence question pairs. The agent performs particularly well on questions that involve straightforward sentence structures. Questions beginning with “Who”, “What”, and “Where” when the relevant information appears directly before or after the main verb. The agent struggles with certain types of cases due to the simplified heuristics used in its reasoning process. For example sentences that contain multiple clauses or unusual word ordering, Questions that require reasoning about relationships, questions with multiple prepositional phrases.

3 HOW EFFICIENT IS YOUR AGENT? HOW DOES ITS PERFORMANCE CHANGE AS THE SENTENCE COMPLEXITY GROWS?

Agent is computationally efficient because it relies on simple rule based heuristics and sequential scanning of the sentence . Most operations performed by the agent involve iterating through the list of tokens in the sentence to identify verbs, subjects, objects, or prepositional phrases. As a result, the time complexity of the main operations is approximately linear in the number of words in the sentence $O(n)$, where n is the number of tokens in the sentence.

When sentences become more complex For example sentences containing multiple verbs, multiple prepositional phrases, or indirect relationships between entities can confuse the heuristic rules. In these situations, the agent may select the wrong phrase as the answer or fail to identify the correct subject or object.

4 DOES YOUR AGENT DO ANYTHING PARTICULARLY CLEVER TO TRY TO ARRIVE AT AN ANSWER MORE EFFICIENTLY?

Agent uses several simple strategies to efficiently identify answers. First it identifying the main verb in the sentence. Once the main verb is found, the agent assumes a simplified structure where words before the verb correspond to the subject and words after the verb correspond to potential objects. This allows the agent to narrow down the search space for possible answers quickly rather than analyzing the entire sentence. The agent analyzes the question to determine the type of information being requested for example “Who”, “What”, “Where”, or “When”. Once the question type is known, the agent searches only for the relevant part of the sentence. For example, “Who” questions cause the agent to look for a subject or person, while “Where” questions cause it to search for prepositional phrases that indicate location.

The agent also uses heuristic rules to guide the search process. such as assuming that the subject appears before the verb and the object appears after the verb, the agent avoids expensive grammatical analysis but still producing correct answers in most cases.

The agent also relies on predefined knowledge, such as lists of names, verbs, pronouns, and prepositions derived from common English words. By categorizing words in advance the agent can quickly determine the grammatical role of each token during processing.

5 HOW DOES YOUR AGENT COMPARE TO A HUMAN? DO YOU FEEL PEOPLE INTERPRET THE QUESTIONS SIMILARLY?

Humans rely on deep linguistic knowledge, contextual understanding, and prior experience when interpreting sentences. In contrast the agent uses simple heuristic rules and pattern matching to extract answers.

For simple sentences with clear grammatical structure, the agent usually arrives at the same answers that a human would. Humans also tend to identify subjects, verbs, and objects when interpreting questions such as “Who brought the note?” or “Where did the horse go?”. In this sense, both the agent and humans rely on recognizing relationships between actions and entities in a sentence.

Human can understand implied meaning, resolve ambiguity, and use common-sense reasoning to infer information that may not be explicitly stated. The agent is limited to extracting information that appears directly in the sentence and can struggle with complex sentences. Humans can also use contextual knowledge and real world understanding, whereas the agent relies only on the rules encoded in code.